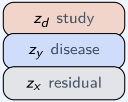
Three DIVA models — domain-invariant VAEs at a glance

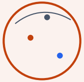
DIVA β-VAE



Split $\mathbf{z}$ *into study, disease & residual.*

- **Backbone:** Gaussian VAE on log-counts.
- Lightest variant — ablation baseline.

DIVA Hyp-PhILR-NB

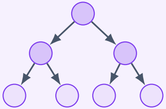


Poincaré ball

The same split, on the Poincaré ball.

- **Backbone:** compositional PhILR (NB).
- Latent on the ball via $\exp_0$ map.

DIVA TreeDTM-VAE



The same split, with a tree likelihood.

- **Backbone:** Dirichlet-tree-multinomial.
- Sibling-centred log-ratio encoder.

Shared DIVA mechanism: latent split $\mathbf{z} = [z_d; z_y; z_x]$ · conditional priors $p(z_d \mid \text{study})$, $p(z_y \mid \text{disease})$ · aux. heads $q(d \mid z_d)$, $q(y \mid z_y)$ · predict from z_y only.